Title: Building a Forecasting Pipeline for Bilateral Trade Flows: Benchmarking Time Series and Machine Learning Models

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Abstract

This project aims to develop a forecasting pipeline for predicting bilateral trade flows at the product level using historical data. Multiple approaches will be implemented and compared, including naive baselines, classical time series models such as ARIMA and exponential smoothing, and machine learning techniques such as gradient boosting and neural networks. The objective is to evaluate their relative performance in an out-of-sample forecasting setting. Additionally, the study examines whether incorporating auxiliary data sources—such as macroeconomic indicators and commodity prices—improves predictive accuracy or introduces noise and overfitting. Model evaluation will be conducted using the Symmetric Mean Absolute Percentage Error (sMAPE), ensuring consistency with established benchmarks, while analyzing trade-offs between model complexity and performance.

Motivation and Problem Statement

Accurate forecasting of trade flows is essential for economic planning, policy-making, and understanding global market dynamics. Reliable predictions support decision-making in areas such as supply chain management, trade policy, and risk assessment. However, traditional econometric models often struggle to capture the high-dimensional, non-linear, and dynamic relationships inherent in disaggregated trade data at the product and country level. In addition, trade flows are frequently affected by disruptive events—such as policy changes, geopolitical tensions, or supply chain shocks—which can cause structural breaks and abrupt shifts in patterns. Standard time series models, which rely on assumptions of stability, may fail to adapt effectively in such contexts. This project addresses these challenges by leveraging modern machine learning techniques alongside classical time series methods. It investigates whether machine learning models, due to their flexibility and ability to model complex non-linear interactions, can better capture both underlying patterns and the effects of disruptive trade shocks. By systematically comparing different approaches in an out-of-sample setting—including periods of structural change—the project aims to identify robust and generalizable forecasting strategies for economic data.

Research Questions

1. How do machine learning models compare to classical time series approaches in forecasting trade flows?
2. Do auxiliary data sources improve forecast accuracy?

3. How do machine learning and classical time series models differ in their ability to capture and adapt to disruptive trade shocks?
4. What is the trade-off between model complexity and generalization performance?

Methodology

The project follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework to ensure a structured, transparent, and reproducible workflow.

Data understanding and exploration will use bilateral trade data from the Observatory of Economic Complexity (OEC) at the HS4 product level. Exploratory Data Analysis (EDA) will include descriptive statistics, correlation analysis, and time series decomposition to identify trends, seasonality, and structural breaks. Particular attention will be given to periods of disruptive events (e.g., policy changes or supply chain shocks). Visualizations such as line plots and heatmaps will help detect patterns, outliers, and inconsistencies.

Data preparation involves cleaning, handling missing values, and transformations such as normalization or log-scaling. Feature engineering will include lag features, rolling statistics, and seasonal indicators. Auxiliary data (e.g., exchange rates, commodity prices, Google Trends) will be temporally aligned and selected carefully. Where appropriate, event-related features (e.g., dummy variables) will be introduced to capture disruptive shocks.

Modeling will cover three groups: (1) naive baselines, (2) classical time series models (ARIMA, SARIMA, ETS), and (3) machine learning methods (e.g., XGBoost, LSTM, transformer-based models). Hyperparameter tuning will be conducted using grid or randomized search.

Evaluation will follow time-aware validation strategies such as rolling or expanding windows. In addition, performance will be assessed around disruptive events by defining pre-shock, shock, and post-shock periods. Models will be evaluated using sMAPE, with comparisons to baseline methods and, where appropriate, statistical tests to assess significance.

Reproducibility will be ensured through version-controlled code and documented preprocessing pipelines.

Expected Results and Evaluation

The expected outcome of this project is the development of a forecasting pipeline for accurately predicting bilateral trade flows at the product level. Machine learning models are expected to outperform naive and classical time series baselines in capturing complex patterns, while the contribution of auxiliary data sources will be critically assessed. Beyond overall predictive accuracy, the project will evaluate model performance during disruptive events and structural breaks. Machine learning models are expected to better capture non-linear adjustments and recovery dynamics following such shocks, although their performance may depend on feature availability and the risk of overfitting to pre-shock patterns. Baseline models (e.g., naive methods, ARIMA, exponential smoothing) will serve as reference points. Performance will be evaluated using the Symmetric Mean Absolute Percentage Error (sMAPE), with MAE as a robustness check, both overall and across event windows (pre-shock, shock, and post-shock periods). Success will be defined as a meaningful improvement over baseline models (e.g., 10–20% reduction in sMAPE), alongside consistent performance across products, countries, time periods, and resilience during periods of disruption.